

Week 8: Testing and Validation Strategies

Testing Plan — Temporal CNN for DDoS Attack Detection

Project	Temporal CNN for DDoS Attack Detection
Dataset	BCCC-Cloud-DDoS-2024 — 700,774 network flows • 86 features • 3 classes (Attack, Benign, Suspicious)
Models	BiLSTM + Soft Attention 1D Residual CNN Soft-Vote Ensemble
Windowing	30 consecutive time-sorted flows per window, stride 15 (50% overlap)

Goal

The goal for Week 8 was to develop a comprehensive testing approach that addresses both **AI model performance** and **security robustness**. This report documents the testing plan that was developed and executed, including the definition of test scenarios, success metrics, relevant tools, and the strategy for AI validation and cybersecurity auditing.

Activities

Model Evaluation and Security Assessment

Both the BiLSTM and 1D Residual CNN models were evaluated on a stratified 15% held-out test set. The test split was created before windowing to prevent data leakage, and an unweighted cross-entropy loss was used for evaluation (as opposed to the class-weighted loss used during training) to ensure a fair, unbiased comparison.

The BiLSTM achieved **93.84% accuracy** and the 1D Residual CNN achieved **94.11% accuracy** on the test set. A detailed per-class classification report was generated using scikit-learn's `classification_report`, producing precision, recall, and F1-score for all three classes to four decimal places. The Attack and Benign classes both achieved F1-scores above 0.94, while the Suspicious class showed a precision of approximately 70%—meaning roughly 30% of Suspicious alerts were false positives. This is a known limitation tied to the minority class imbalance.

A 3×3 confusion matrix was computed for each model and visualized as annotated heatmaps using Seaborn. The matrices were plotted side-by-side and revealed that Suspicious flows are most commonly confused with Attack flows—a less dangerous error from a security standpoint, as it represents over-alerting rather than missed threats.

Training and validation loss curves, along with accuracy curves, were plotted epoch-by-epoch for both models in a 2×3 subplot grid. These curves were monitored for overfitting (train/validation divergence). Both models converged without significant divergence, confirming that the anti-overfitting measures—dropout, batch normalization, label smoothing, and gradient clipping—were effective. Early stopping with a patience of 7 epochs was employed, and both models stopped training before reaching the maximum of 30 epochs.

A soft-vote ensemble was implemented to combine both architectures. The ensemble extracts softmax probability matrices from both models on the test set and averages them element-wise. Soft voting was chosen over hard voting because it accounts for model confidence—a highly confident prediction from one model can outweigh a hesitant prediction from the other. The ensemble's accuracy and macro-F1 were compared against both individual models, and it achieved performance at or above both, with a narrowed gap on the Suspicious minority class.

For the security assessment dimension, a confidence-based escalation analysis was conducted to answer a key operational question: *when should the IDS auto-classify traffic versus escalate to a human analyst?* The maximum softmax probability was computed for each test window from the ensemble, and a sweep of 40 thresholds (from 0.50 to 0.99) was performed. At each threshold, accuracy on high-confidence windows and overall coverage were measured. Three visualizations were produced: a confidence distribution histogram, a coverage-versus-accuracy trade-off curve, and a per-model threshold comparison. The results demonstrated that higher thresholds yield higher accuracy at the cost of lower coverage, providing a principled basis for setting an operational escalation policy.

Test Data Gathering and Test Case Documentation

The primary test data was derived from the BCCC-Cloud-DDoS-2024 dataset through a carefully designed pipeline that ensures temporal integrity and prevents data leakage. The following measures were implemented:

- **Chronological sorting:** All 700,774 flows were sorted by their original timestamp before any processing, preserving real temporal ordering of network traffic.
- **Scaler fitted on training time only:** The MinMaxScaler was fit exclusively on the first 70% of chronologically sorted rows. The full dataset was then transformed using this scaler, so no future statistical information leaked into the normalization.
- **Feature selection on training time only:** Both the variance threshold filter and correlation-based pruning (reducing 324 features to 86) were fitted on training-time rows only.
- **Stratified window-level split:** After building 30-flow sliding windows, the dataset was split into train (70%), validation (15%), and test (15%) using stratified sampling to ensure proportional representation of the Suspicious minority class across all splits.

- **No window bridging:** The split was performed at the window level after windowing, so no individual window spans the boundary between training and test data.

Milestone / Deliverable: Testing Plan

A. Definition of Test Scenarios

#	Test Scenario	Description	Result
1	Holdout Test Evaluation	Evaluate BiLSTM and 1D CNN on a stratified 15% held-out test set using unweighted loss for fair comparison	BiLSTM 93.84%, CNN 94.11%
2	Per-Class F1 Report	Generate precision, recall, and F1 for Attack, Benign, and Suspicious individually to 4 decimal places	Attack/Benign F1 $\geq$ 0.94; Suspicious precision $\approx$ 70%
3	Confusion Matrix	Compute 3×3 confusion matrices and visualize as annotated heatmaps to reveal misclassification patterns	Suspicious $\rightarrow$ Attack dominant misclass.
4	Overfitting Detection	Plot training vs. validation loss and accuracy curves epoch-by-epoch for both models; monitor for divergence	No significant overfitting; early stopping triggered
5	Ensemble Validation	Combine BiLSTM and CNN via soft-vote (probability averaging); compare accuracy and macro-F1 against individual models	Ensemble $\geq$ both; Suspicious gap narrowed
6	Confidence Threshold Analysis	Sweep 40 confidence thresholds (0.50–0.99) on ensemble; measure accuracy vs. coverage trade-off for escalation policy	Higher threshold = higher accuracy, lower coverage
7	Data Leakage Prevention	Verify temporal sorting, scaler fitted on training-time only, feature selection on training-time only, stratified splits with no window bridging	No leakage; all checks passed

B. Success Metrics

Metric	Target	Achieved	Met?
Overall Test Accuracy	$\geq$ 93%	93.84% (BiLSTM), 94.11% (CNN)	Yes
Attack / Benign F1	$\geq$ 0.94	$\geq$ 0.94 for both classes	Yes

Suspicious Precision	$\geq 75\%$	$\approx 70\%$	Near (known limitation)
No Overfitting	Train/val curves converge	Confirmed; early stop triggered	Yes
Ensemble $\geq$ Individual Models	Ensemble matches or exceeds	Ensemble $\geq$ both on macro-F1	Yes
Data Leakage	Zero future leakage	Scaler/selector fit on train only	Yes

C. Relevant Tools

Tool	Purpose	How It Was Used
PyTorch	Model training, inference, and evaluation	Forward pass on test set, softmax extraction for ensemble and confidence analysis
scikit-learn	Classification metrics and data splitting	classification_report, confusion_matrix, f1_score, accuracy_score, train_test_split (stratified)
Matplotlib / Seaborn	Visualization of results	Loss/accuracy curves, confusion matrix heatmaps, confidence distribution plots, coverage-accuracy curves
NumPy	Numerical computation	Probability averaging for ensemble, threshold sweep computation, NaN/Inf handling
MinMaxScaler	Feature normalization	Fitted on training-time rows only to prevent data leakage
VarianceThreshold	Low-variance feature removal	Fitted on training-time rows; reduced feature set as part of leakage-free pipeline

D. Strategy for AI Validation

The AI validation strategy was built around a multi-layered evaluation approach. Rather than relying on a single accuracy number, the strategy evaluated model quality from several complementary angles:

- **Holdout evaluation with unweighted loss** ensured that the reported accuracy reflects true generalization rather than the effect of class weighting used during training.

- **Per-class breakdown** revealed that aggregate accuracy can mask poor performance on minority classes—specifically, the Suspicious class’s ~70% precision was not visible from the overall 93–94% accuracy alone.
- **Confusion matrix visualization** identified the specific direction of misclassification (Suspicious → Attack), which informed a risk assessment that this error mode is safer than the reverse.
- **Training/validation curve monitoring** confirmed that the models are not overfitting and that anti-overfitting measures (dropout, BatchNorm, label smoothing, gradient clipping, early stopping) are working as intended.
- **Ensemble comparison** validated that combining architecturally different models (sequence-based BiLSTM and pattern-based CNN) through soft voting improves robustness on the hardest class.
- **Confidence threshold sweep** translated model output probabilities into an actionable escalation policy, bridging the gap between model evaluation and real-world deployment.

E. Strategy for Cybersecurity Auditing

The cybersecurity auditing strategy focused on ensuring that the IDS pipeline itself is trustworthy and that its outputs can be relied upon in a security-critical context. The following areas were addressed:

Data Integrity Assurance

The most critical security concern for an ML-based IDS is data integrity—if the training pipeline leaks future information or is contaminated, the model’s reported performance is unreliable. The pipeline was designed with explicit leakage prevention at every stage: chronological sorting before processing, scaler and feature selectors fitted only on training-time data, and stratified splits at the window level. These measures ensure that the model’s test-set performance reflects genuine generalization.

Confidence-Based Escalation for Operational Security

The confidence threshold analysis directly addresses a cybersecurity concern: an IDS that auto-classifies every flow with equal certainty is a liability. By establishing a threshold-based escalation policy, flows below the confidence threshold are flagged for human analyst review rather than auto-classified. This reduces the risk of both false negatives (missed attacks) and false positives (unnecessary alerts) by routing uncertain decisions to human judgment.

Misclassification Risk Assessment

The confusion matrix analysis served as a security risk assessment. The finding that the dominant misclassification direction is Suspicious → Attack (over-alerting) rather than Attack → Benign (missed attacks) is a favorable outcome from a security perspective. Over-alerting creates analyst fatigue but does not create security blind spots. The ~70% Suspicious precision was documented as a known limitation, with the ensemble and confidence thresholding serving as mitigations.

Planned Security Assessments

For the remaining weeks of the project, additional security testing is planned to harden the system. This includes vulnerability scanning of any exposed inference endpoints using tools such as Nmap and OWASP ZAP, model file integrity verification using SHA-256 checksums on saved .pt and .pkl artifacts, and adversarial robustness testing using FGSM/PGD perturbations via IBM's Adversarial Robustness Toolbox (ART) to measure how the models respond to crafted adversarial inputs.

Conclusion

The Week 8 testing plan has been developed and partially executed. On the AI validation side, all seven test scenarios have been completed: holdout evaluation, per-class F1 analysis, confusion matrix analysis, overfitting detection, ensemble validation, confidence threshold analysis, and data leakage prevention verification. On the cybersecurity side, the data integrity and confidence escalation strategies have been implemented, with vulnerability scanning and adversarial robustness testing planned for subsequent weeks.

The key results confirm that both models perform well on the 3-class DDoS detection task (93–94% accuracy), with the soft-vote ensemble offering the best overall balance. The primary area for continued improvement remains the Suspicious class precision (~70%), and the confidence threshold analysis provides a practical mechanism for managing this limitation through human-in-the-loop escalation.

#	Test Scenario	Key Result	Status
1	Holdout Test Evaluation	BiLSTM 93.84%, CNN 94.11%	Complete
2	Per-Class F1 Report	Attack/Benign F1 ≥ 0.94 ; Suspicious precision $\approx 70\%$	Complete
3	Confusion Matrix Analysis	Suspicious $\rightarrow$ Attack is dominant misclassification	Complete
4	Overfitting Detection	No divergence; early stopping triggered	Complete
5	Ensemble Validation	Ensemble $\geq$ both; Suspicious class gap narrowed	Complete
6	Confidence Threshold Analysis	Escalation policy informed by coverage-accuracy curve	Complete
7	Data Leakage Prevention	All pipeline stages verified; zero future leakage	Complete